Trayectoria

$x = [0 \text{ a } 10]$

$f(x) = 2\sin(x)e^{0.3x}$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = 2\sin(t)e^{0.3t} \qquad \dot{y}(t) = 2(\cos(t)e^{0.3t} + e^{0.3t}\,0.3\sin(t))$

condiciones iniciales

$x(0) = 0$

$y(0) = 2\sin(0)e^{0.3(0)} = 0$

$\theta(0) = \tan^{-1}(2(\cos(0)e^{0.3(0)} + e^{0.3(0)}\,0.3\sin(0)))$

$= \tan^{-1}(2)$

$$v(t) = \sqrt{1 + (2(\cos(t)e^{0.3t} + e^{0.3t}\,0.3\sin(t)))^2}$$

$$\theta = \tan^{-1}(2(\cos(t)e^{0.3t} + e^{0.3t}\,0.3\sin(t)))$$

$$w(t) = \frac{d(\tan^{-1}(2(\cos(t)e^{0.3t} + e^{0.3t}\,0.3\sin(t))))}{dt} = \frac{2(-0.91e^{0.3t}\sin(t) + 0.6e^{0.3t}\cos(t))}{1 + (2(\cos(t)e^{0.3t} + e^{0.3t}\,0.3\sin(t)))^2}$$

$2(-0.91e^{0.3t}\sin(t) + 0.6e^{0.3t}\cos(t))$

Trayectoria 2

Tramo 1

$x = [0 \leq x < 3$

$f(x) = 0.5x^2$

condiciones iniciales

$x(0) = 0$

$y(0) = 0$

$\theta(0) = \tan^{-1}(0)$

$x(t) = t \qquad \dot{x}(t) = 1$

$y(t) = 0.5t^2 \qquad \dot{y}(t) = t$

$$v(t) = \sqrt{1 + t^2}$$

$$\theta(t) = \tan^{-1}(t)$$

$$w(t) = \frac{d(\tan^{-1}(t))}{dt} = \frac{1}{1 + t^2}$$

1

Tramo 2

$x = [3 \leq x < 7]$

$f(x) = [-(x-5)]^2 + 8$

$x(t) = t \qquad \dot{x}(t) = 1$

$f(t) = -(t-5)^2 + 8 \qquad \dot{y}(t) = -2t + 10$

$$v(t) = \sqrt{1 + (-2t+10)^2}$$

$$w(t) = \frac{d(\tan^{-1}(-2t+10))}{dt} = \frac{-2}{1 + (-2t+10)^2}$$

-2

$$\theta = \tan^{-1}(-2t + 10)$$

Tramo 3

$x = [7 \leq x \leq 10]$

$f(x) = 2e^{(x-7)} \cos(x-7)$

$x(t) = t$ $\qquad$ $\dot{x}(t) = 1$

$y(t) = 2e^{(t-7)} \cos(t-7)$ $\qquad$ $\dot{y}(t) = 2\left(e^{t-7} \cos(t-7) - e^{t-7} \sin(t-7)\right)$

$$v(t) = \sqrt{1 + \left(2\left(e^{t-7} \cos(t-7) - e^{t-7} \sin(t-7)\right)\right)^2}$$

$$\theta(t) = \tan^{-1}\left(2\left(e^{t-7} \cos(t-7) - e^{t-7} \sin(t-7)\right)\right)$$

$$\omega(t) = \frac{d\left(\tan^{-1}\left(2\left(e^{t-7} \cos(t-7) \overset{-4e^{x-7}\sin(x-7)}{- e^{t-7} \sin(t-7)}\right)\right)\right)}{dt} = \frac{-4e^{t-7} \sin(t-7)}{1 + \left(2\left(e^{t-7} \cos(t-7) - e^{t-7} \sin(t-7)\right)\right)^2}$$